



KAZAKH-BRITISH
TECHNICAL
UNIVERSITY



APPROVED

by Dean of SAM

T.Kenzhebayev

«13» january 2026

SYLLABUS

Academic year 2025-2026

Semester: Spring 2026

Discipline: *Discrete Structures*

Volume of credits: 3 (2/0/1)

Course/Syllabus designer: *Kalmurzayev Birzhan*

Instructor's e-mail	d.nurlanbek@kbtu.kz
Class Schedule	According to the schedule, wsp.kbtu.kz
Office Hours	TBA
Office	311

1. Course Overview

Discrete Structures is a transfer-oriented course designed to meet the requirements of Computer and Information Science degree programs. The course provides an introduction to the mathematical foundations essential for computer science, including logic, set theory, relations, functions, combinatorics, graph theory, and etc.

2. Aims and objectives

This course is designed to meet the following objectives of the program. Upon successful completion of this course, students will be able to:

- Demonstrate critical thinking, analytical reasoning, and problem-solving skills
- Apply appropriate mathematical concepts and operations to interpret data and to solve problems
- Identify a problem and analyze it in terms of its significant parts and the information needed to solve it.
- Formulate and evaluate possible solutions to problems and select and defend the chosen solutions.

3. Learning outcomes

At the end of the course, students are expected to:

- create compound statements, expressed in mathematical symbols or in English, to determine the truth or falseness of compound statements and to use the rules of inference to prove a conclusion statement from hypothesis statements by applying the rules of propositional and predicate calculus logic;
- prove mathematical statements involving numbers by applying various proof methods, which are based on the rules of inference from logic;
- prove the validity of sequences and series and the correctness of repeated processes by applying mathematical induction;
- define and identify the terms, rules, and properties of set theory and use these as tools to support problem solving and reasoning in applications of logic, functions, number theory, sequences, counting, trees and graphs, and finite state machines;
- solve recursive problems by applying knowledge of recursive sequences;
- create graphs and trees to represent and help prove or disprove statements, to make decisions or select from alternative choices, to calculate probabilities, to document derivation steps, or to solve problems; and
- construct and analyze finite state automata (another name for machines), formal languages, and regular expressions.

Knowledge: during the study of this course, students must obtain knowledge about how to: explain with examples the basic terminology of functions, relations, and sets; perform the operations associated with sets, functions, and relations; convert logical statements from informal language to propositional and predicate logic expressions; apply formal methods of symbolic propositional and predicate logic, such as calculating validity of formulae and computing normal forms; Identify the proof technique used in a given proof; determine which type of proof is best for a given problem; explain the parallels between ideas of mathematical and/or structural induction to recursion and recursively defined structures; explain the relationship between weak and strong induction and give examples of the appropriate use of each; state the well-ordering principle and its relationship to mathematical induction; apply counting arguments, including sum and product rules, inclusion-exclusion principle and arithmetic/geometric progressions; apply the pigeonhole principle in the context of a formal proof; be familiar with elementary concepts of Languages and Automata theory and understand their role in Computer Science.

Skills: As a result of studying this course, students must be able to

- apply formal logic proofs and/or informal, but rigorous, logical reasoning to real problems, such as predicting the behavior of software or solving problems such as puzzles;
- Demonstrate comprehension of discrete structures and their relevance within the context of computer science, in the areas of data structures and algorithms, in particular;
- Apply discrete structures into other computing problems such as formal specification, verification, databases, artificial intelligence, and cryptography;
- Demonstrate mathematical skills, analytical and critical thinking abilities;
- Communicate clearly and effectively using the technical language of the field correctly.

4. Textbook and readings

Primary textbooks

1. Kenneth Rosen: Discrete Mathematics and Its Applications, 7th edition, 2012, McGraw-Hill. Press, 2009.
2. Susanna S. Epp: discrete mathematics with applications, 5th edition, 2018 Cengage.

3. Ding-Zhu Du, Ker I-Ko. Problem solving in Automata, Languages and Complexity. 2001, John Willey & Sons, Inc.

Supplementary textbooks

1. Ralph P. Grimaldi: Discrete and Combinatorial mathematics: an applied introduction. 3rd edition, 1994, Addison-Wesley Publishing Company, Inc.
2. Seymour Lipschutz, Marc Lipson: Discrete Mathematics. (Shaum's Outline), 4th edition, 2009, McGraw-Hill.

5. Lesson Program

COURSE CALENDAR

Week	Class work				Learning Objectives
	Topic	Lecture classes	Seminar Classes	Chapters for Reading	
1	Introduction to Set Theory: Operations, Relations, Functions	2	1	Lecture notes and [1, Chapters 1–2]	- Understand sets, subsets, operations - Learn relations and types - Understand functions and mappings
2	Combinatorial Principles and Proof Techniques: Mathematical Induction, Counting Principles	2	1	Lecture notes and [1, Chapters 4–5]	- Apply principles of counting - Learn and apply mathematical induction - Solve combinatorial problems
3	Binomial Coefficients and Combinatorial Counting: Factorials, Permutations, Combinations, Functions on Finite Sets	2	1	Lecture notes and [1, Chapters 5–6]	- Compute permutations, combinations - Understand binomial coefficients - Apply combinatorial methods to finite sets
4	Fibonacci Numbers and Recurrence Relations	2	1	Lecture notes and [1, Chapter 6]	- Define and compute Fibonacci numbers - Solve linear recurrence relations - Apply recurrences to counting problems
5	Generating Functions and Applications in Combinatorics	2	1	Lecture notes and [1, Chapter 6-7]	- Understand ordinary generating functions - Apply generating functions to solve recurrences - Model combinatorial problems using generating functions
6	The Euclidean Algorithm and Modular Arithmetic: Applications to Diophantine	2	1	Lecture notes and [1, Chapter 7-8]	- Compute GCD using Euclidean algorithm - Solve linear Diophantine equations - Apply Chinese Remainder

	Equations, Chinese Remainder Theorem, Euler's Theorem				Theorem and Euler's Theorem
7	Propositional Logic: Boolean Functions, Equivalences, Complete Sets of Functions, Normal Forms (CNF, DNF)	2	1	Lecture notes and [2, Chapter 1-3]	- Understand Boolean functions - Apply equivalence transformations - Construct CNF and DNF representations
8	Predicate Logic: Predicates, Quantifiers, Free and Bound Variables, Formalization of Statements	2	1	Lecture notes and [2, Chapter 3-4]	- Understand predicates and quantifiers - Distinguish free and bound variables - Formalize mathematical statements
9	Introduction to Graph Theory: Basic Concepts and Graph Types	2	1	Lecture notes and [1, Chapter 9-10]	- Define graphs, vertices, edges - Classify types of graphs - Understand basic properties of graphs
10	Graph Theory: Connectivity, Trees, and Graph Isomorphism	2	1	Lecture notes and [1, Chapter 10-11]	- Analyze connectedness - Understand trees and spanning trees - Study graph isomorphism problem
11	Introduction to Automata Theory: Regular Languages, Regular Expressions, Graph Representation of RE	2	1	Lecture notes and [3, Chapter 1]	- Define regular languages - Use regular expressions - Represent automata as graphs
12	Finite Automata: Deterministic (DFA) and Nondeterministic (NFA)	2	1	Lecture notes and [3, Chapter 2]: 2.1, 2.2., 2.3	- Construct DFA and NFA - Simulate automata computations - Understand differences between DFA and NFA
13	Conversion of NFA to DFA, Minimization of DFA	2	1	Lecture notes and [3, Chapter 2]: 2.5., 2.6, 2.7	- Convert NFA to equivalent DFA - Minimize DFA - Apply minimization to optimize automata
14	Context-Free Grammars (CFG)	2	1	Lecture notes and [3, Chapter 3]: 3.1, 3.2	- Define CFG and derivations - Generate strings using CFG - Understand applications of CFG in parsing
15	Pushdown Automata (PDA) and Context-Free Languages	2	1	Lecture notes and [3, Chapter 3]: 3.4, 3.5	- Define PDA - Recognize context-free languages using PDA - Relate CFG and PDA

6. Course requirements and grades

COURSE ASSESSMENT PARAMETERS

Attendance	0%
Homeworks	10%
Quiz 1 & Quiz 2	20%
Midterm/endterm	30%
Final exam	40 %
Total	100%

№	Assessment criteria	Weeks																Total	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 17		
1.	Attendance has to be more than 70%																		
2.	Homework		1	1		1	1		1	1	1		1	1		1		10	
3.	Quizes and mid/end-terms				10			15				10			15			50	
4.	Final examination																40	40	
	Total	60																40	100

Grading policy:

Intermediate attestations (on 7th and 14th week) join topics of all lectures, homework, quiz and materials for reading discussed to the time of attestation. The maximum number of points within attendance, activity, homework and quiz for each attestation is 30 points. Every intermediate attestation and quiz are held during practice hours in the university.

Final exam joins and generalizes all course materials, is conducted in the complex form with questions and problems. Final exam duration is 150 min. Maximum number of points is 40. At the end of the semester, you receive an overall total grade (summarized index of your work during semester) according to conventional KBTU grade scale.

Grade		Achievement percentage	Assessment criterion
«Excellent»	A	95-100%	This grade is given when the student: demonstrated a complete understanding of the course material; did not make any errors or inaccuracies;

	A -	90-94%	completed control and laboratory work in a timely and correct manner, and submitted reports on them; demonstrated original thinking; submitted control quizzes on time and without any errors; completed homework assignments; engaged in research work; independently used additional scientific literature in studying the discipline; was able to independently systematize the course material.
«Good»	B+	85-89%	This grade is given when the student: Has mastered the course material at no less than 75%; Did not make gross errors in responses; Timely completed control and laboratory work and submitted them without fundamental remarks; Correctly completed and timely submitted control tests and homework assignments without fundamental remarks;
	B	80-84%	
	B-	75-79%	
	C+	70-74%	Utilized additional literature as indicated by the instructor; Engaged in research work, made non-fundamental errors, and fundamental errors corrected by the student themselves; Managed to systematize the course material with the help of the instructor.
«Satisfactory»	C	65-69%	This grade is given when the student: Has mastered the course material no less than 50%; Required assistance from the instructor when completing control and laboratory work, homework assignments; Made inaccuracies and non-fundamental errors when submitting control tests; Did not demonstrate activity in research work, relied solely on the educational literature indicated by the instructor; Experienced more difficulty in systematizing the material.

7. General Rules

Academic Integrity

Students are required:

- to be respectful to the teacher and other students;
- to switch off mobile phones during classes;
- DO NOT cheat. Plagiarized papers shall be graded with zero points!

- to come to classes prepared and actively participate in classroom work; to meet the deadlines;
- to enter the room before the teacher starts the lesson;
- to attend all classes. No make-up tests or quiz are allowed unless there is a valid reason for missing it;
- to follow KBTU academic policy regarding W, AW, I, FX, F grades.
- When students are absent for 30% of the lessons or more, then their grade is F.
- When students have a score of 29 or less for attestation 1 added to attestation 2, then their grade is F.
- When students have a score between 9,5 and 19,4 for their final exam, then their grade is FX.
- When students have a score of 9,4 or less (less for their final exam, then their grade is F.
- When students do not come for their final exam, then their grade is F.

Students are encouraged to:

- consult the teacher on any issues related to the course;
- make up within a week's time for the works undone for a valid reason without any grade deductions.

Attendance policy

Activity and attendance in lessons is mandatory. Mandatory requirement is preparation for each lesson. When students are absent for 30% of the lessons or more, then their grade is F.

Course Instructor:



Dias D. Nurlanbek

Approved by the meeting of the SAM

Meeting number: #5

Date: «13» january 2026